Activity 01 - Use AI to Clean Data and Extract Insights

- Course: Infographics and Data Visualization with PowerPoint (TGS-2026064178)
- Duration: 35 minutes
- Tools: Approved AI assistant + Excel + PowerPoint
- Output: Annotated insight slide, editable PPTX and completed checklist

Objective

Use ai to organise and clean approved mock sources, then verify and convert candidate patterns into defensible business insights.

Scenario

Northstar Retail's marketing lead has a one-page customer survey infographic built from 420 synthetic Q2 responses. Leadership wants to know which customer experience deserves further investigation. Use an approved AI assistant with the supplied CSV, workbook and source notes to produce a clean analysis and candidate insights. Verify every number in Excel before recommending one next step.

Detailed instructions

1. Register the sources. Open the A01 XLSX, CSV and Source Notes sheet. In the AI Prompt Lab record each source, its period, data status, owner and permitted use before uploading anything.

2. Preserve the raw data. Save a named working copy. Do not modify or replace the raw CSV; perform cleanup on a separate worksheet or in an approved AI session.

3. Run the data-cleaning prompt. Copy the Data analyst prompt from PROMPTS.md. Ask the AI to flag duplicates, missing values, inconsistent labels, units and periods and to describe every proposed transformation. Require no invented values.

4. Verify the cleanup. Compare the AI response with the raw CSV. Apply only justified changes in the workbook, retain a before/after note and confirm the response count remains 420.

5. Ask for candidate insights. Request the largest, smallest, gap, exception and competing explanation. Require each candidate insight to cite the exact rows or source fields used.

6. Reproduce the calculations. Use the Analysis sheet and Excel formulas to reproduce every critical difference or percentage. Reject any AI claim that cannot be reproduced.

7. Write the insight chain. Complete What it shows; What it may mean; Why it matters; What action is justified. Add one caveat or confidence note.

8. Build the PowerPoint evidence. Open the A01 start PPTX, save a working copy, add one editable annotation and show Synthetic Northstar Retail data, Q2 2026, n=420 in the source footer.

9. Complete the prompt log. Record the AI tool, full prompt, approved inputs, output reviewed, rejected claims, human changes, reviewer and final owner.

Suggested timing

- 5 minutes - Review the question and evidence definitions
- 8 minutes - Read values and denominators

- 12 minutes - Build the insight chain and annotation
- 7 minutes - Peer challenge and revision
- 3 minutes - Save and submit

Included inputs and assets

- Synthetic CSV and formula-enabled XLSX mock data
- Editable PowerPoint start and worked-example files
- PROMPTS.md with copy-ready data-analysis, infographic, Gemini Notebook and text-to-image prompts
- AI Prompt Lab worksheet for provenance and human-change records

Submission

Submit the editable PPTX, completed XLSX, source CSV, prompt/source log, used assets and checklist. Record every AI tool, prompt, source, rejected claim and material human change.

Debrief

The final insight clearly separates what the data shows, what it may mean, why it matters and what should happen next.